

Summary of Classical Mendelian Genetics

Classical Mendelian genetics, founded on Gregor Mendel's pea plant experiments, explains how traits are inherited through discrete units called genes. Each gene has alternative forms called alleles, which determine an organism's genotype and phenotype. Dominant alleles mask recessive ones in heterozygotes.

Key Principles

- **Law of Segregation:** Two alleles for a trait separate into different gametes during meiosis (anaphase I), resulting in a 3 : 1 phenotypic ratio in monohybrid crosses.
- **Law of Independent Assortment:** Alleles of different genes assort independently into gametes due to random chromosome alignment at metaphase I, producing a 9 : 3 : 3 : 1 phenotypic ratio in dihybrid crosses.

Tools for Prediction and Analysis

- **Punnett squares** predict offspring genotypes and phenotypes and their ratios.
- **Pedigrees** trace inheritance patterns across generations, distinguishing autosomal dominant and recessive traits.
- **Probability rules** simplify calculations for crosses involving multiple independently assorting genes.

Extensions Beyond Simple Dominance

- **Incomplete dominance:** Heterozygotes show an intermediate phenotype (e.g., pink snapdragons).
- **Codominance:** Both alleles are fully expressed separately (e.g., human ABO blood groups).
- **Sex linkage:** Traits on sex chromosomes, often X-linked recessive, are more common in males due to their single X chromosome.

Biological Basis

Mendel's laws are explained by chromosome behavior during meiosis:

- Segregation occurs when homologous chromosomes separate at anaphase I.
- Independent assortment results from random alignment of chromosome pairs at metaphase I.

Importance

These foundational concepts and tools form the basis of modern genetics, enabling prediction and understanding of inheritance patterns in organisms.